Problem 1.[10 points]

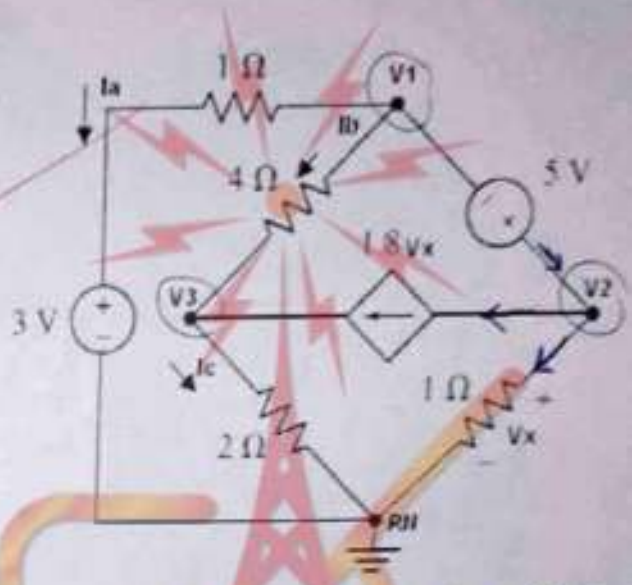
In the circuit of figure. Use nodal analysis method to find the nodal voltages and branch currents:

1- Write the required nodal equations (don't simplify)

(1) $\frac{V_1 - V_3}{5} + 1.8V_x = \frac{V_3}{2}$ ..... $5 = 4$

(2) $I_a + I_a = \frac{V_2 - 0}{1}$ ..... (couldn't simplify)

(3) $1.8V_x + \frac{V_1 - V_3}{4} + \frac{V_2}{1}$



2- Write the expressions for the branch currents:

Ia= $\frac{V_1 - 3}{1}$ ..... $= V_1 - 3 = -4.647 - 3 = -1.647$ A

Ib= $\frac{V_1 - V_3}{4}$ ..... $= -0.98575$

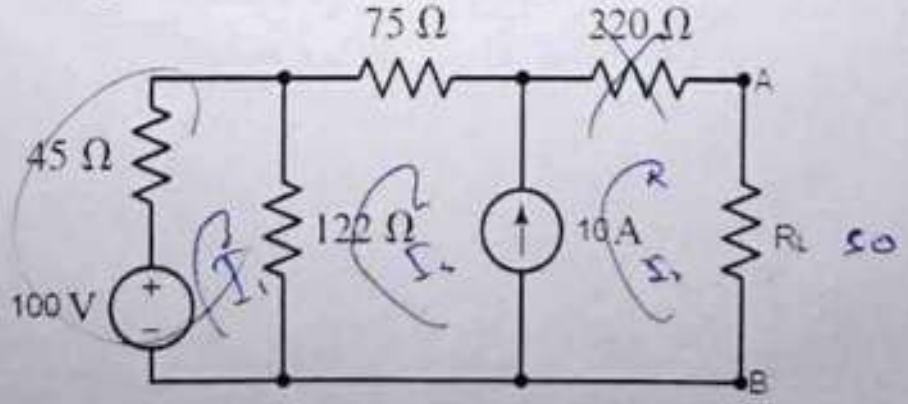
Ic= $\frac{V_3 - 0}{2}$ ..... $= \frac{-0.704}{2} = -0.3522$

Problem 2.[10 points]

For the circuit shown in figure find:

1- The Thevenin equivalent circuit with respect to terminals A-B.

2- The current through load resistance (R_L=50 Ω).



Good Luck

Al Balqa' Applied University
Faculty of Engineering Technology
Electrical Engineering Department

كلية الهندسة التكنولوجية

Student Name: [redacted]

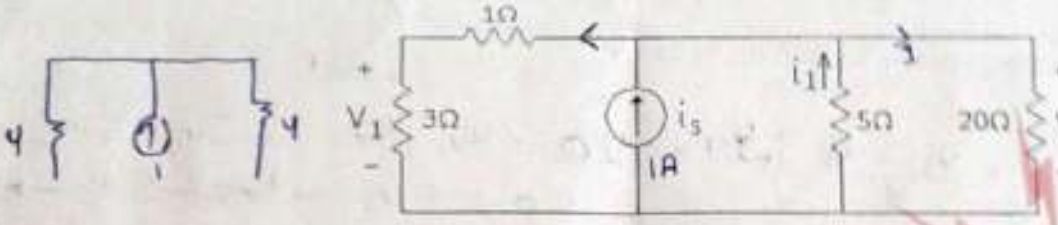
First Semester: 2013/2014
First Exam

#spark_team

3171502006 2

Course Name: Electrical Circuits (1)	Date: 2/11/2013
Instructors: Dr. Dwairi, Dr Abu Harb, Eng. Juma Abu-Ghalune	Exam Time: 50 Minutes

Problem 1: Consider the circuit shown where $i_s = 1A$.
Write the equations that are used to solve the problem



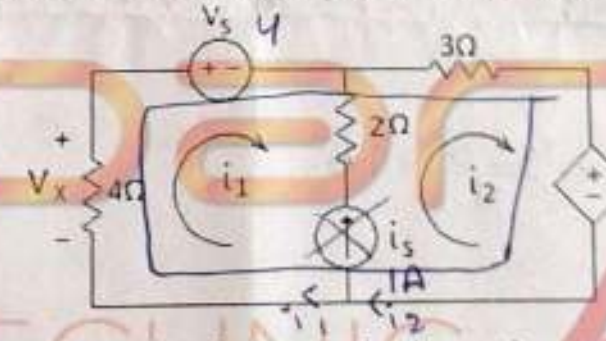
(2 // 5) → 4Ω
(1, 3) → 4Ω
$i_1 = \frac{1 * 4}{8} = \frac{1}{2}$

a- Using current divider rule, find i_1 and v_2

$i_1 = \frac{5 * 20}{25} = .4A$, $v_2 = 20 * .1 = 2V$

b- Using voltage divider rule find v_1. $v_1 = \frac{2 * 3}{4 \; 2} = \frac{3}{2}V$

Problem 2: Consider the circuit shown where $i_s = 1A$, and $v_s = 4$ V.



a- Using mesh analysis, write all the required equations to find i_1, i_2 and v_x

supermesh

1- $4i_1 + 3i_2 + 4 + 3v_x = 0 \rightarrow (1)$

$-8i_1 + 3i_2 = -4$

$i_2 = 1 + i_1 \rightarrow (2)$

2-

$v_x = -4i_1 \rightarrow (3)$

3-

$i_1 = 1.4A$, $i_2 = 2.4A$, $v_x = -4i_1 = -5.6V$

b- Find the power supplied by the dependent source

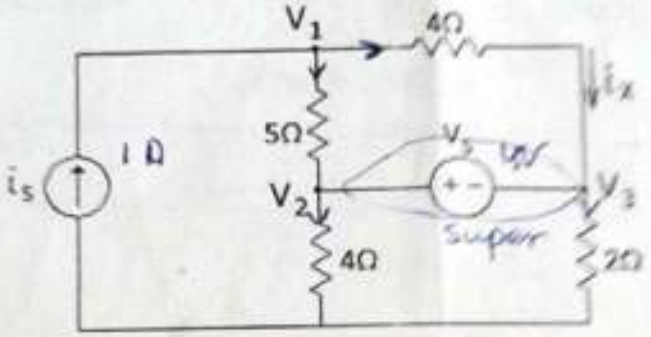
$P = \frac{v^2}{R} \Rightarrow \frac{(3 * -5.6)^2}{3} = 94.08 w$

$P = vi = 3 * -5.6 * -2.4$

$= +40.32 w$

Problem 3:

Solve the circuit shown using nodal method where $i_s = 1A$, and $v_x = 4V$,



- Write all the required equations to find v_1, v_2, v_3

node 1

$$1 = \frac{v_1 - v_2}{5} + \frac{v_1 - v_3}{4} \quad \times 20$$

$$20 = 4v_1 - 4v_2 + 5v_1 - 5v_3$$

$$20 = 9v_1 - 4v_2 - 5v_3 \rightarrow (1)$$

super node

$$\frac{v_1 - v_2}{5} + \frac{v_1 - v_3}{4} + \frac{v_3}{2} + \frac{v_2}{4} = 0 \rightarrow 9v_1 - 9v_2 - 5v_3 = 0$$

$$v_2 - v_3 = 4$$

$v_1 =$ 4V, $v_2 =$ 4V, $v_3 =$ 0V

- Find i_x

$$i_x = \frac{v_1 - v_3}{4} = \frac{4 - 0}{4} = 1A$$

$= 1A$

Spark POLYTECHNIC

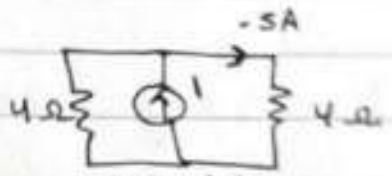
First semester 2013/2014 electric circuit-
2/11/2013 First exam

problem 1 $j_s = 1A$

$(5 // 20) = 4\ \Omega$

$j_1 = \frac{-.5 * 20}{25} = -.4A$

عكس اتجاه المفروض



يتوزع تيار على فرعين متساويين لأنهما متساويين

$v_2 = .4 * 5 = 2V$

or $i_{20} = .5 - .4 = .1\ A$

$v_{20} = .1 * 20 = 2V$

v_1 by voltage divider $= \frac{2 * 3}{4} = \frac{3}{2} = 1.5V$

problem 2 $j_s = 1A$, $v_s = 4V$

a) $-4j_1 + v_s + 3j_2 + 3v_x = 0$ but $v_s = 4V$

$-4j_1 + 3i_2 + 3v_x = -4$

2- $j_2 = j_1 + 1$

3- $v_x = -4j_1$

بحل 3 معادلات

$j_1 = 1.4A$ $j_2 = 2.4A$ $v_x = -4i_1 = -5.6V$

b) $p = vj = -9.4 * 3 * -5.6 = 40.32$ watt

problem 3 $j_s = 1A$, $v_s = 4V$

a) 1- $1 = \frac{v_1 - v_2}{5} + \frac{v_1 - v_3}{4}$

2- $v_2 - v_3 = 4$

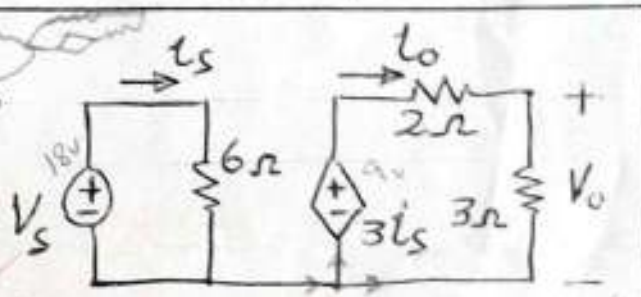
3- $\frac{v_1 - v_2}{5} + \frac{v_1 - v_3}{4} = \frac{v_3}{2} + \frac{v_2}{4}$ | $v_1 = 4V$ $v_2 = 4V$ $v_3 = 0V$

b) $i_x = \frac{v_1 - v_3}{4v} = \frac{4-0}{4}$

Electric Circuit I
First Exam

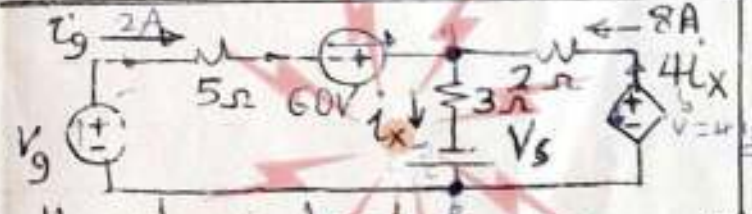
Summer 2012/2013
23/6/2013

1 Consider the circuit shown, where $V_s = 18V$. Find i_s, i_o, and V_o.



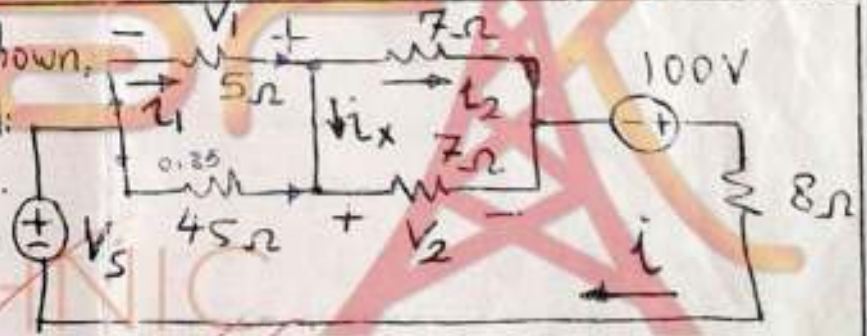
$i_s = 3A$, $i_o = 1.8A$, $V_o = 5.4V$

2 For the circuit shown, when $i_g = 2A$ find:
a. i_x , V_s , V_g.
b. the power supplied by the dependent source.



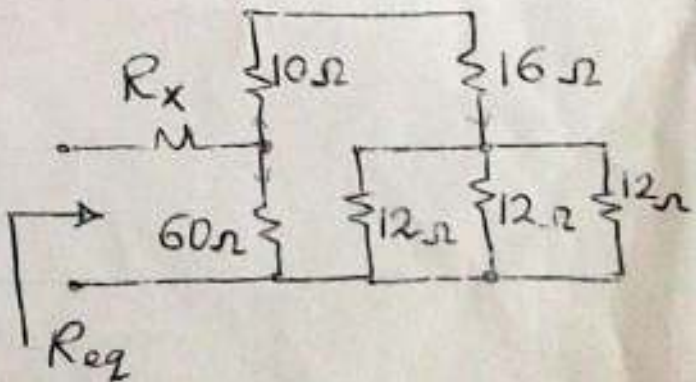
$i_x = 10A$, $V_s = 54V$, $V_g = -74V$, $P = 320$ watt

3 Consider the circuit shown, where $V_s = 36V$. Find:
i , V_1 , V_2 , i_1 , i_2 , i_x.



$i = 8.5A$, $V_1 = 32.25V$
$V_2 = 29.75V$, $i_1 = 0.65A$
$i_2 = 4.25A$, $i_x = 3.4A$

4 For the circuit shown, find R_{eq}, when $R_x = 20\ \Omega$.



$R_{eq} = 40\ \Omega$.

Summer 2012/2013 electric circuit 1

23/6/2013 First exam

Q1 $i_s = \frac{v_s}{6} = 8A$ $-3i_s + 2i_o + 3i_o = 0$ $i_o = \frac{3}{5} i_s = 1.8A$

$-3i_s + 2i_o + 8v_o = 0$ $v_o = 5.4V$

Q2 $i_g + 8A = i_x$ $i_x = 10A$ because $i_g = 2A$

$-4i_x + 2x_8 + 3i_x - v_s = 0$ $v_s = 6V$

$-v_g + 5*i_g = 60 + 3*i_x - v_s = 0$

$v_g = -26V$

$P = Vi = 8 * 4 * i_x = 320 watt$

Q3 $v_s = 36V$ $(45\Omega // 5) = 4.5\Omega$ $(7 // 7) = 3.5\Omega$

$-36 + 4.5i + 3.5i - 100 + 8i = 0$ $i = \frac{136}{16} = 8.5A$

$v_{4.5} = v_1 = i * 4.5 = 8.5 * 4.5 = 38.25V$

$v_2 = v_{3.5\Omega} = 8.5 * 3.5 = 29.75V$

$i_1 = \frac{38.25}{5} = 7.65A$ $i_2 = \frac{29.75}{7} = 4.25A$

$i_x = 3.4A$

Q4 $(12 // 12 // 12) = 4\Omega$ $(4, 16\Omega) = 20\Omega$

$(10\Omega, 10\Omega) = 20\Omega$ $(30\Omega // 60\Omega) = 20\Omega$

$R_{eq} = 20 + 20 = 40\Omega$

Al Balqa' Applied University
Faculty of Engineering Technology
Electrical Engineering Department

كلية الهندسة التكنولوجية

Student Name

Summer Semester: 2013/2014
First Exam

Course Name: Electrical Circuits (1)
Instructors: Dr I. Abu-Harb and Eng J. Abu-Ghalune

Date: June 29, 2014
Exam Time: 45 mins.

Prob.1: Choose the correct answer.

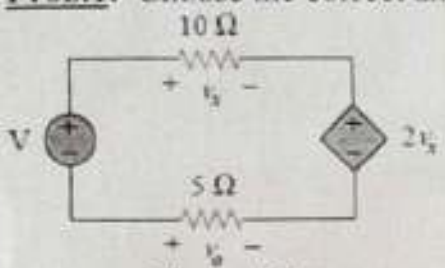


Figure B.1

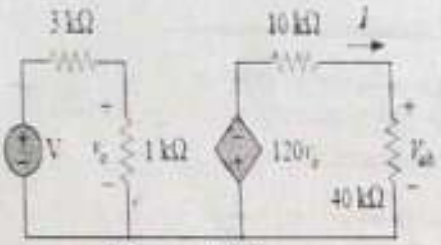


Figure B.2

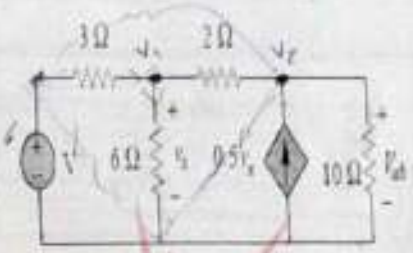


Figure B.3

Consider the circuit shown in Fig. B.1 where V = 7 V.

1. The value of the current in the circuit I is
   a. 0.3 A b. 0.1 A c. 0.2 A d. 0.4 A e. None of all
2. The value of the voltage V_x is
   a. 5 V b. 2 V c. 3 V d. 1 V e. None of all
3. The power supplied by the dependent source is
   a. – 3.2 W b. – 1.8 W c. – 0.4 W d. – 0.8 W e. None of all

Consider the circuit shown in Fig. B.2 where V = 6 V.

4. The value of the voltage V_o is
   a. 0.75 V b. 3 V c. 1.5 V d. 2.25 V e. None of all
5. The value of the voltage V_{ab} is
   a. – 54 V b. – 36 V c. – 18 V d. –72 V e. None of all
6. The value of the current in the circuit I is
   a. – 1.8 mA b. 7.2 mA c. –0.2 mA d. – 5.4 mA e. None of all

Consider the circuit shown in Fig. B.3 where V = 6 V.

7. The value of the voltage V_x is
   a. 12 V b. 18 V c. 24 V d. 6 V e. None of all
8. The value of the voltage V_{ab} is
   a. 40 V b. 10 V c. 30 V d. 20 V e. None of all
9. The power absorbed by the 2-Ω resistor is
   a. 32 W b. 50 W c. 8 W d. 128 W e. None of all

ePark POLYTECHNIC

Student Name: .....

Prob. 1	1	2	3	4	5	6	7	8	9
Answer	c	b	c	c	c	c	b	d	c

8

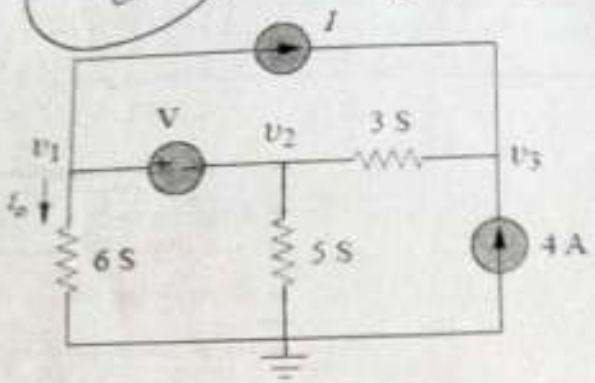


Figure B.4

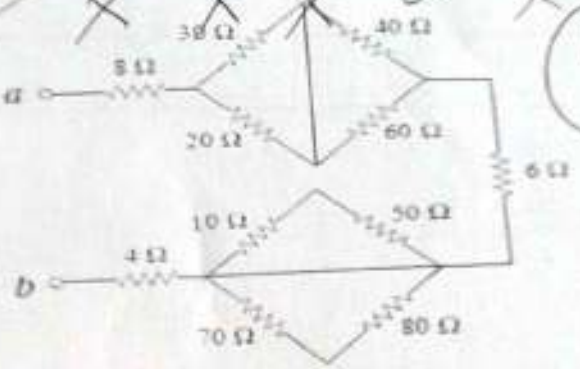


Figure B.5

Prob. 2: Consider the circuit shown in Fig. B.4 where V = 6 V and I = 1.5 A. Use nodal method in analysis.

a. Write all the required equations to calculate V_1, V_2, V_3, and i_o.

1. $i_o = 6V_1$

2. $V_1 - V_2 = 6$

3. $5V_2 + 8(V_2 - V_3) + 6V_1 = -1.5$ (i_o)

4. $3(V_3 - V_2) + 5V_2 = 4$

b. Solve the equations to find V_1 = 2.54 V, V_2 = −3.159 V, V_3 = 3.49 V, and i_o = 17.04 A

Prob.3: Find the equivalent resistance between the *a-b* terminals for the circuit shown in Fig. B.5.

Solution :

$30 \parallel 20$

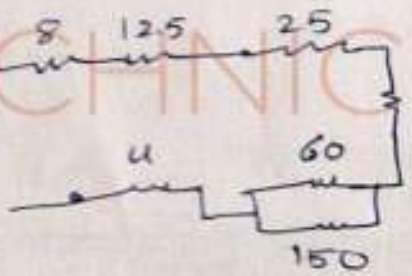
$= 12$ (5)

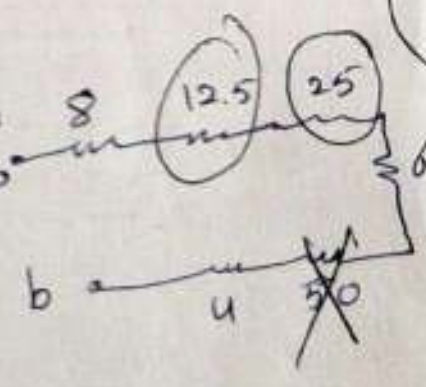
$40 \parallel 60$

$= 25$

$60 \parallel 150$

$= 50$





$8 + 12.5 + 25 + 6 + 4 + 50$

$R_{eq} = 105.5\ \Omega$

Summer semester 2013/2014 — electric circuit(1) First exam — June 29, 2014 (B)

prob.1

1. $-7 + 10I + 2V_x + 5I = 0$ but $V_x = 10I$

   so $I = \frac{7}{35} = .2A$ ©

2. $V_x = 10I = 2V$ (b)

3. $P = vi = -.2 * 2 * 2 = -.8W$ (a)

* consider the circuit show in B.2 $V = 6V$

4. $-6 + 4I = 0$ $I = \frac{6}{4} = 1.5mA$ $v_o = 1 * 1.5 = 1.5V$ ©

5. $120v_o + 10I + 40I = 0$ but $v_o = 1.5$ $I = -\frac{180}{50} = -3.6mA$

   $v_{ab} = -3.6 * 40 = -144V$ (e)

6. $I = -3.6mA$ (e)

* Consider the circuit shown in Fig B.3 $V = 6V$

7. by using nodal $\frac{6 - V_x}{3} = \frac{V_x}{6} + \frac{V_x - V_{ab}}{2}$

   and $\frac{V_x - V_{ab}}{2} + .5V_x = \frac{V_{ab}}{10}$ by solving eqn

   $V_x = 12V$ (a)

8. $V_{ab} = 20V$ (d)

Spark POLYTECHNIC

9. $P = \frac{V^2}{R} = \frac{(V_2 - 20)^2}{2} = 32$ watt ②

prob 2 $V = 6V$ $I = 1.5A$ use nodal method

a. 1- $6V_1 + 5V_2 + I = 0$ but $I = 1.5A$

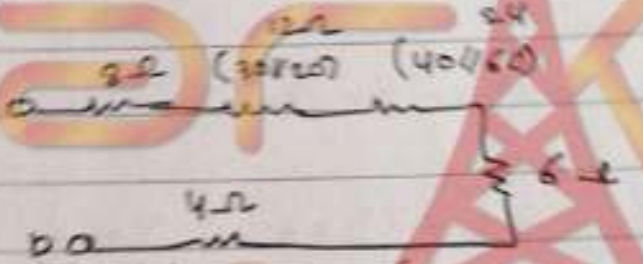
$6V_1 + 5V_2 = -1.5A - (V_2 - V_3) * 3$

2- $V_1 - V_2 = 6$

3- $4 + 1.5 + 3(V_2 - V_3) = 0$

b $V_1 = 3.09V$ $V_2 = -2.91V$ $V_3 = -1.075V$ $i_o = 18.9A$

prob 3 Req



Req = 3.4 Ω

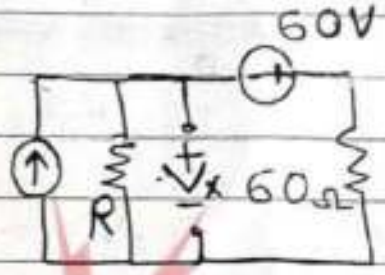
المقاومة 6Ω في آخر السلك لتوضيح السلك فارغ

Electric Circuit I

Second exam.
14/7/2013

Write all the required equations that are used to solve the following problems.

1 Using superposition principle, find V_x when $I_s = 8\,A$, $R = 30\,\Omega$.



a. V_x due to voltage source only.

Eq. $\frac{-V_x}{30} = \frac{V_x + 60}{60} \rightarrow -3V_x = 60$

$V_x = -20$ volt



$V_x = -20$ volt

b. V_x due to current source only.

Eq. $8 = \frac{V_x}{30} + \frac{V_x}{60}$

$3V_x = 480$

$V_x = 160$ volt



$V_x = 160$ volt

c. V_x due to both sources.

Eq. $V_{x\,total} = V_x + V_x$

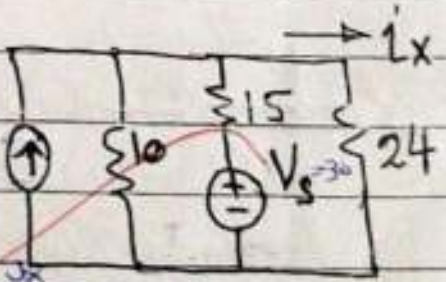
$-20 + 160 = 120$ volt

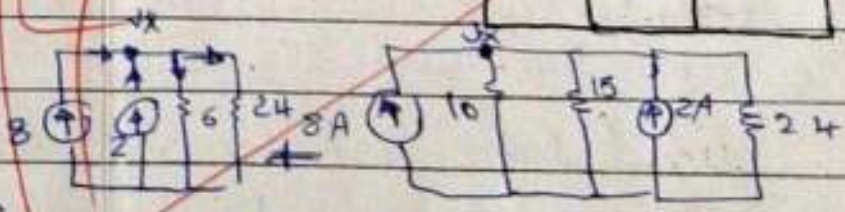
$V_x = 120$ volt

2 Using source transformation method, find i_x.

$V_s = 30V$, $I_s = 8\,A$.

Circuit





$\frac{8+2}{1} = \frac{V_x}{6} + \frac{V_x}{24}$

$10 = \frac{5V_x}{24}$

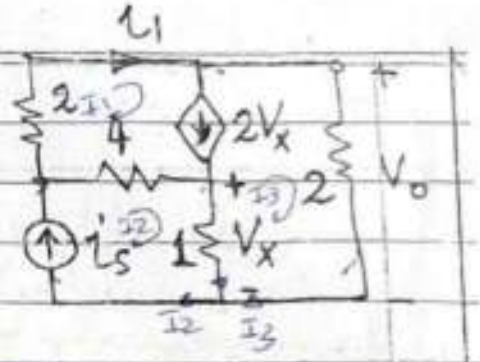
$V_x = 48$

$i_x = \frac{V_x}{24} = \frac{48}{24} = 2A$

Eq. $i_x = 2A$

$i_x = 2A$

3 Using mesh method, find i_1 and V_o when $i_s = 8$ A.



$(i_x = I_2 - I_3)$ $V_x = i_x(1)$

1) $I_1 - I_3 = 2V_x$ → $I_1 + I_3 - 2I_2 = 0$

Super mesh Loop

2) $2I_1 + 2I_3 + 1(I_3 - I_2) + 4(I_1 - I_2) = 0$ → $6I_1 - 5I_2 + 3I_3 = 0$

3) $I_2 = 8$ A

$i_1 = -2.66$ A

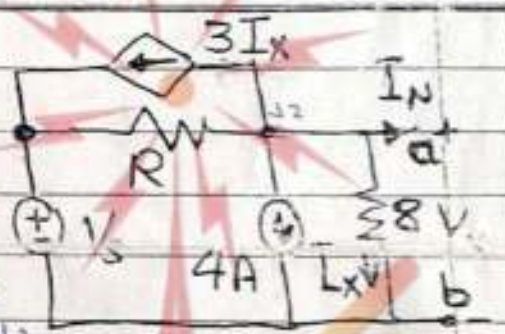
V_o

4) $6I_1 + 3I_3 = 40$, $I_1 + I_3 = 16$ ⟹ $I_1 = 2.66$ A, $I_3 = 18.6$ A

$V_o = 37.2$ volt

$V_o = 2I_3 \rightarrow 2(18.6) = 37.32$

4 Consider the circuit shown where $R = 4\ \Omega$, $V_s = 20$ V.



a. Using nodal method find V_x.

at node 1 $V_1 = 20$ volt $\{ i_x = \frac{V_2}{8}$

at nod 2 $I_2 = 4 + 3\left(\frac{V_2}{8}\right) + \frac{V_2}{8}$

$\frac{V_1 - V_2}{4} = 4 + \frac{4V_2}{8}$

$\frac{20 - V_2}{4} = \frac{4V_2}{8} + 4$

$5 - \frac{V_2}{4} = \frac{4V_2}{8} + 4$ $\quad 5 - 4 = \frac{4}{8}V_2 + \frac{V_2}{4} \Rightarrow 1 = \frac{3V_2}{8} \rightarrow V_2 = \frac{8}{3} = 2.6$

$V_x = V_2$

$V_x = 2.66$ volt

b. Short circuit the a-b terminals ($V_x = 0$) then using mesh method find I_N.

$I_1 - I_2 = 4$

$-20 + 4(I_1 - I_4) + 8(I_2 - I_3) = 0$

$8(I_3 - I_2) = 0$

$I_4 = 3(I_2 - I_3) = 0$

$I_4 - 3I_2 + 3I_3 = 0$

$I_4 = 3I_x$

$I_x + I_3 = I_2$

$I_x = I_2 - I_3$

$I_3 = I_N$

$I_N = 1$ A

c. Find the Thevenin's equivalent resistance R_{th} between the a-b terminals.

$I_1 - 3I_2 + 3I_3 = 0$

$4I_1 + 4I_2 - 8I_3 = 0$

$-3I_2 + 8I_3 = -1$

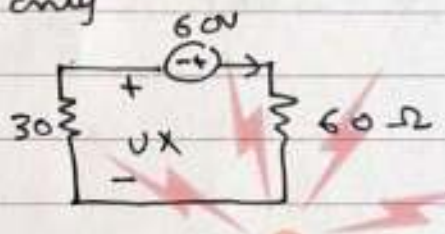
$R_{th} = \frac{1}{0.235}$

$R_{th} = 4.255\ \Omega$

Summer 2012/2013 electric circuit 1
14/7/2013 Second exam

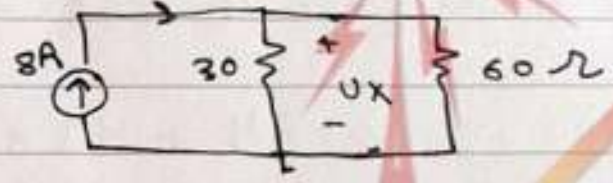
Q1 $I_s = 8A$, $R = 30\Omega$ Find v_x

a. v_x due the voltage source only



$$v_x = -\frac{60 * 30}{90} = -20V$$

b. v_x due the current source only



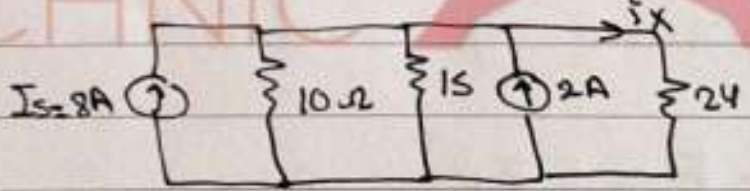
$$v_x = \frac{8 * 60}{90} * 30 = 160V$$

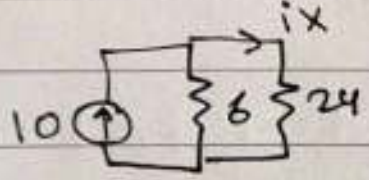
c. v_x due to both sources

$$v_x = -20 + 160 = 140V$$

Q2 Find i_x $v_s = 30V$ $I_s = 8A$

حول v_s و I_s الى مصدر تيار وقاوومة على التوازي





$$i_x = \frac{10 * 6}{30} = 2A$$

Q3 Find i_1 and v_o when $i_s = 8A$

1) $6i_1 + 3i_3 - 5i_2 = 0$
2) $i_2 = 8A$
3) $i_1 = 2v_x + i_3 \Rightarrow i_1 = 2i_2 - i_3$
4) $v_x = 1(i_2 - i_3)$

$i_1 = -2.66A$ $i_3 = 18.66A$ $v_o = 2i_3 = 37.33V$ وبحل المعادلات

Q4 $R = 4\Omega$, $V_s = 20V$ Find v_x by nodal

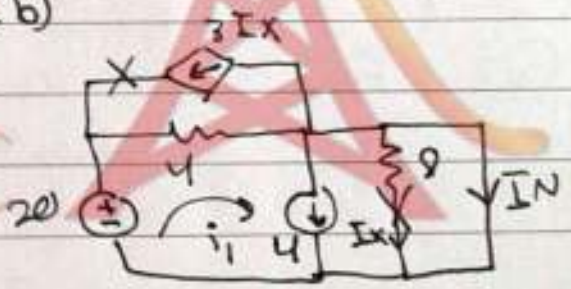
$V_1 = V_s = 20V$

node 2 $\quad 4 + 3I_x + I_x = \frac{20 - v_x}{4}$ but $V_x = 8I_x$

$4 + 4I_x = 5 - 2I_x$

$6I_x = 1 \quad I_x = .166A \quad V_x = 1.33$

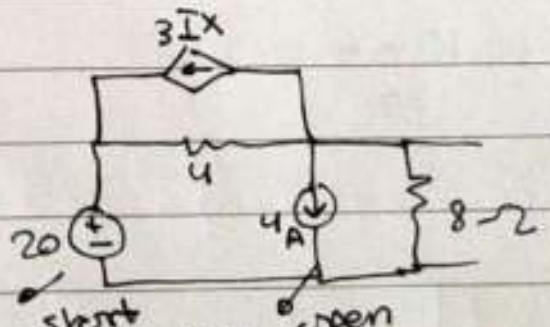
b. Find I_N between (a b)



~~$-20 + 4I_1$~~ $\quad I_N = \frac{20}{4} = 5A$

because $I_x = 0$

c. Find R_{th}



$3I_x + I_x + I = 1$

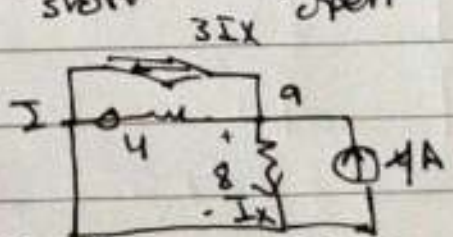
$4I_x + I = 1$

$-4I + 8I_x = 0$

$I = 2I_x$

$I_x = .5IA$

$I_x = \frac{1}{6} \quad v = 8 * \frac{1}{6} = 1.33 \quad R_{th} = 1.33\Omega$



Al Balqa' Applied University
Faculty of Engineering Technology
Electrical Engineering Department

كلية الهندسة التكنولوجية

Student Name: ...

First Semester: 2013/2014
First Exam

Course Name: Electrical Circuits (1)	**Date:** Dec. 14, 2014
Instructors: Dr. M. Dwairi, Dr. I. Abu Harb, Eng. J. Abu-Galune	**Exam Time:** 50 minutes

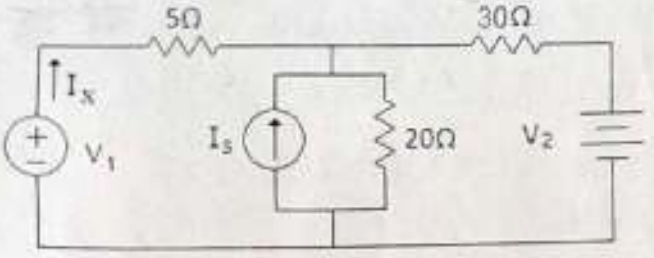


Figure A.1

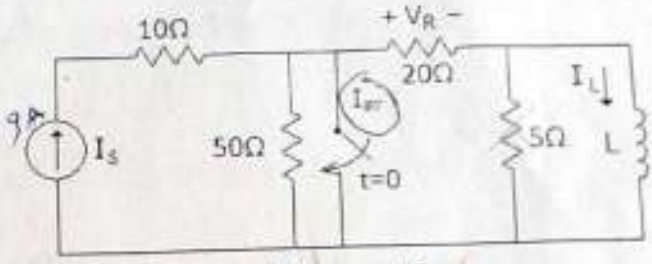


Figure A.2

Problem 1: Using superposition method, find $\mathbf{I_x}$ for the circuit shown in Fig. A. 1where I_s = 3 A, V_1 = 6V, and V_2 = 8V according the following steps:

a. Due to the left voltage source $\mathbf{V_1}$:

$\mathbf{I_{x_1}} = \frac{V}{R_{eq}} = \frac{6}{17} = 0.35\ A$

b. Due to the current source $\mathbf{I_s}$:

$\mathbf{I_{x_2}} = -I_s \times \frac{R_{eq}}{R_{eq}+R_{20}} = -3 \times \frac{5}{17} = -2.11\ A$

c. Due to the right voltage source $\mathbf{V_2}$:

$\mathbf{I_{x_3}} = \frac{V}{R_{eq}} = \frac{8}{34} = 0.18 = I_x$

$I_x = \frac{-8}{34} \times \frac{20}{25}$

d. Due to the all sources:

$\mathbf{I_x} = I_{x_1} + I_{x_2} + I_{x_3} = 0.35 + -2.11 + 0.18 = -1.94\ A$

Problem 2: Consider the circuit shown in Fig. A.2 where the switch is opened for a long time and closed at t = 0. Assume L = 0. 5 H and I_s = 3A, then complete:

a. For t < 0:

$I_L = \frac{3 \times 5}{7.5} = 2.14\ A$

b. For t > 0:

1. The time constant (τ) = $\frac{L}{R_{eq}} = \frac{0.5}{4} = \frac{1}{8}\ s$
2. $I_L = I_{L(0)} e^{-t/\tau} = 2.14 e^{-8t}$
3. $I_{sw} = \ldots$

$V_L = V_R = L \frac{di}{dt} = 0.5 \times 2.14 \times -8 e^{-8t} = -8.56 e^{-8t}\ V$

$I_{sw} = \frac{V_R}{20} = \frac{-8.56 e^{-8t}}{20} = -0.428 e^{-8t}\ A$

Problem 3: Consider the circuit shown in Fig. A.3 where the switch is opened for a long time and closed at t = 0. Assume C = 5 F and I_s = 3 A, then complete :

a. For t < 0:

1. V_R = ..2.X.20..... = ...40V.....

2. The energy stored in the capacitor W_C = ..$\frac{1}{2}$..C..V_o^2 = $\frac{1}{2} \times 5 \times (40)^2$ = 4000 J

b. For t > 0:

1. The time constant (τ) = ..RC = ... = ...4X5 = 20s

2. V_R = Vc = $V_o e^{-t/\tau}$ ..... = 40 $e^{-0.05t}$ ..... V .....

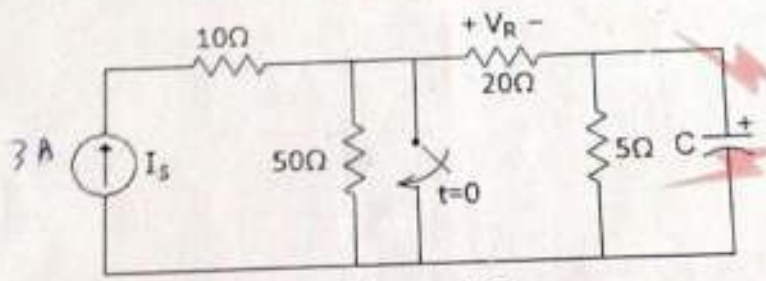


Figure A.3

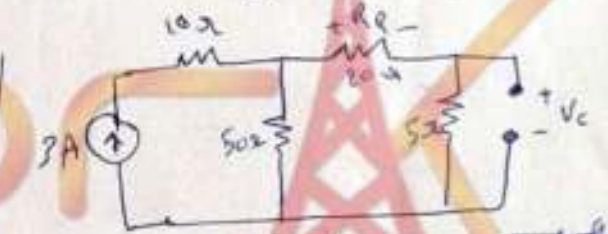
b) t > 0



$R_{eq} = \frac{20 \times 5}{25} = 4\Omega$

$V_R = V_c = V_o e^{-t/\tau} = 40 e^{-\frac{1}{20}t}$

t < 0





a)

$i = 3 \times \frac{50}{75} =$ 2 A

$V_R = i \times 20 =$

$= 2 \times 20 = 40$ V

SPA POLYTECHNIC

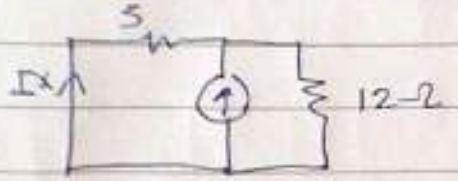
First semester 2013/2014 — Electrical circuit I second exam — 14/Dec/2014 — A

problem 1

a. Due to the left voltage source V1

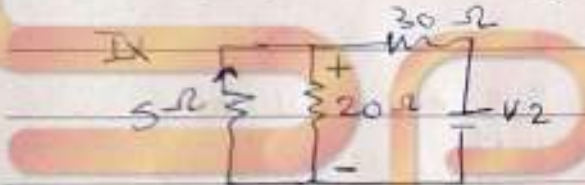
$(30 // 20) = 12\ \Omega$ $\quad I_x = \frac{6}{12+5} = 0.353A$

b. Due to the current source Is



$I_x = \frac{-3 \times 12}{12+5} = -2.1176\ A$ (عكس اتجاه)

C. Due to the right voltage source V_2



$V_{20\Omega} = V_{5\Omega} = \frac{8 \times 4}{34} = .941\ V$

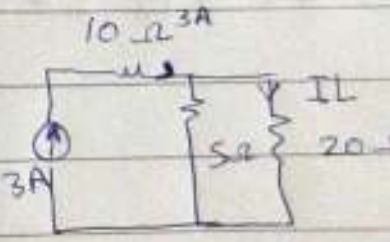
$I_x = -\frac{.941}{5} = -.188\ A$ (اتجاه التيار عكس اتجاه الرسم)

d. Due to the all sources

$I_x = I_{x1} + I_{x2} + I_{x3} = 0.35 - 2.1176 - .188 = -1.95$

problem 2

a. For t<0 $\quad I_L = \frac{3 \times 50}{70} = 2.14A$

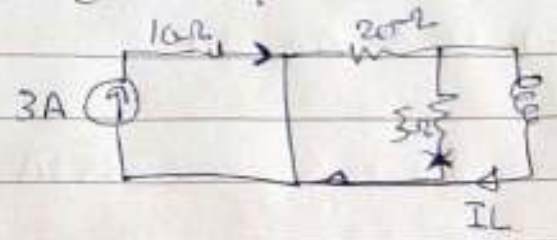


POLYTECHNIC

b. For $t > 0$

1. The Time constant $(\tau) = \frac{L}{R_{eq}} = \frac{.5}{20//5} = .125\,S$

3. I_{sw} ...

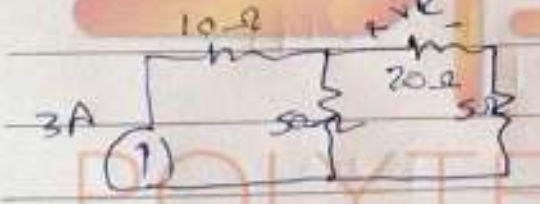


... $I_{sw} = 3 - I_L = 3 - 2.14 = .86\,A$

2. $I_L = I_{L(0)}\, e^{-\frac{t}{\tau}} = 2.14\, e^{-8t}$

problem 3

a. For $t < 0$



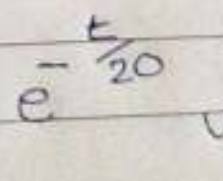
$V_R = i_R * R = \frac{3 * 50}{50+20+5} * 20 = 40V$

2. The energy stored in capacitor $E = \frac{1}{2} C V_0^2 = \frac{1}{2} * 5 * (10)^2 = 25$

b. For $t > 0$

1. The Time constant $(\tau) = RC = (20//5) * 5 = 20\,S$

2. $V_R = -V_0\, e^{-\frac{t}{\tau}} = -10\, e^{-\frac{t}{20}}$



(∵ VC ...

Q1: Find the voltage (v) for the network shown in Fig 1 using superposition theorem. (10marks)

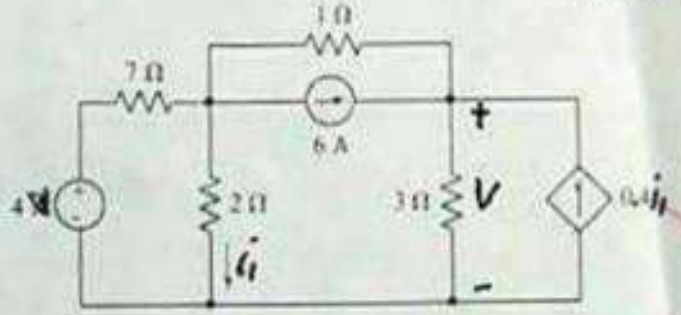


Fig 1

Q2: For the circuit shown in Fig 2: (10marks)

a) Determine the value of Rl to which a maximum power can be delivered .

b) The maximum power deliverd to Rl.

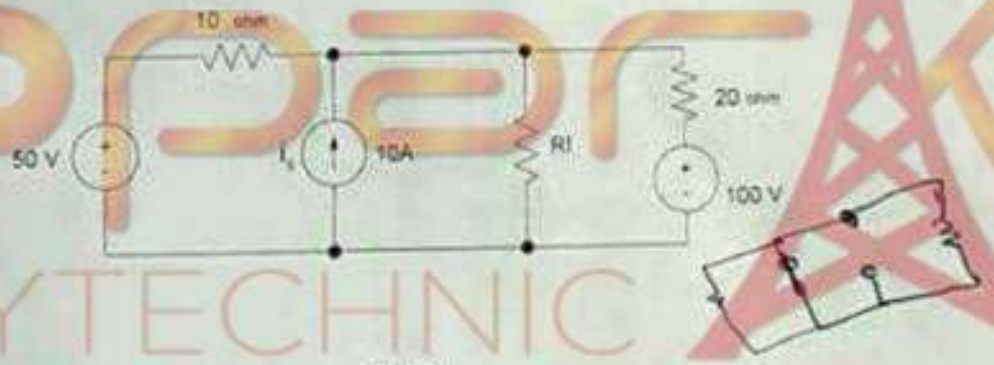


Fig 2

Q3: a) For the circuit shown in Fig 3.a find the equivalent capacitance (10 marks)

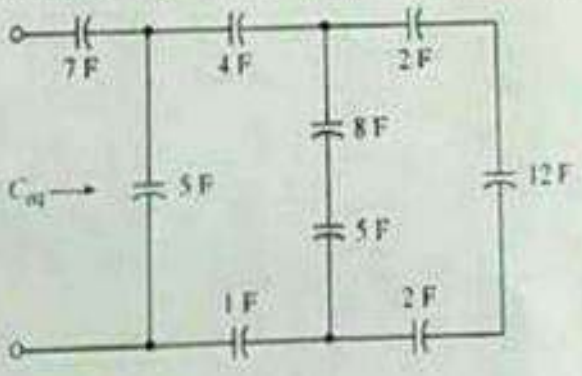
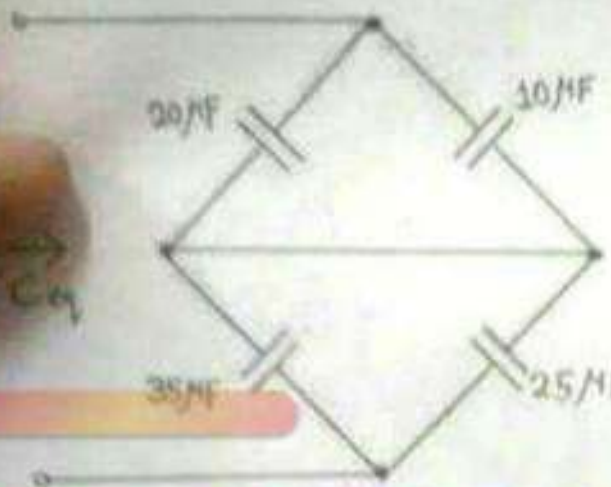


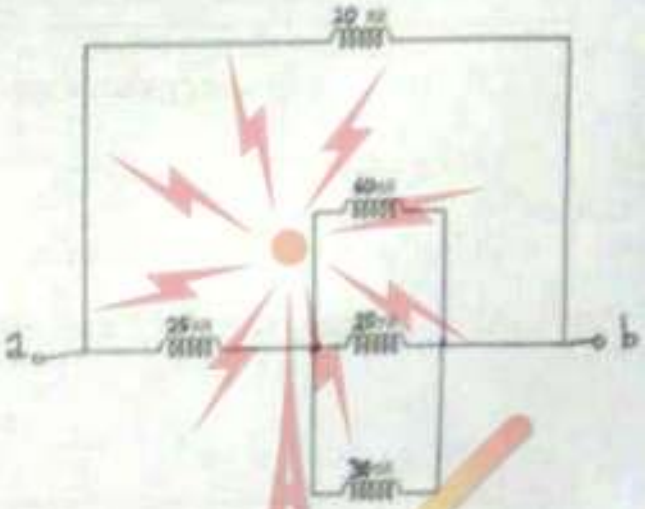
Fig 3.a

1) For the previous circuit, Determine the Thevenin and Norton equivalents for the given network.

2) Find the resistance that dissipate maximum power if it's connected between a and b terminals.

Q4: For the given circuits, calculate Ceq and Leq .





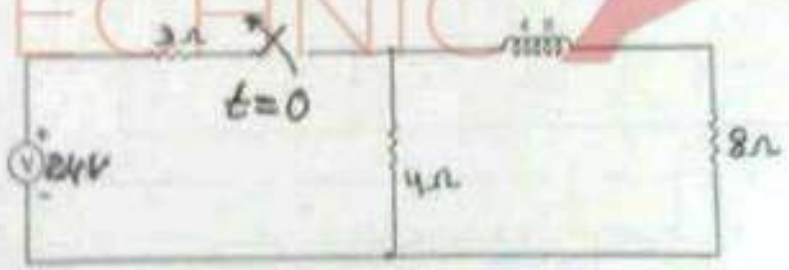
Q5: For the given circuit below find the current in the inductor at:

1) t= 0⁻ (t < zero).

2) t= ∞

3) t= 2s

4) The energy dissipated in 8Ω resistor.



End of examinations papers.

Dr.Yousif Eltous --- Dr. Ibrahim Abu Harb

Al-Balqa' Applied University
Faculty of
EngineeringTechnology
Electrical circuits 1

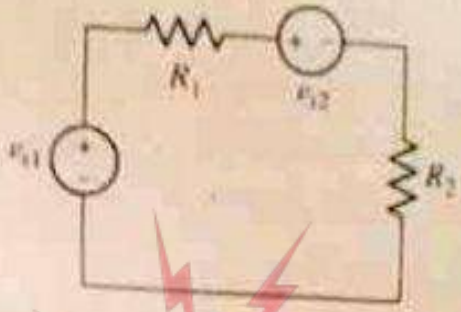
جامعة البلقاء التطبيقية
كلية الهندسة التكنولوجية

First exam 25-06-2015

Problem 1.[9 points]

In the circuit of figure, V_{s1} =100V, V_{s2}=40V, R_1=20Ω, and R_2=10Ω.Compute:

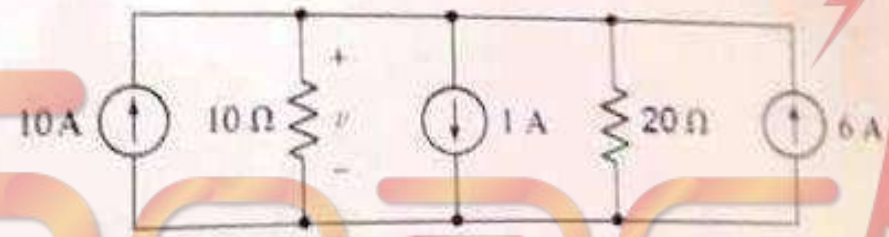
1- The voltage across R_1 (V_{R1}).
2- The current through the circuit.
3- The power absorbed by V_{s1} and V_{s2}.
4- The power absorbed by R_2.



Problem 2.[7 points]

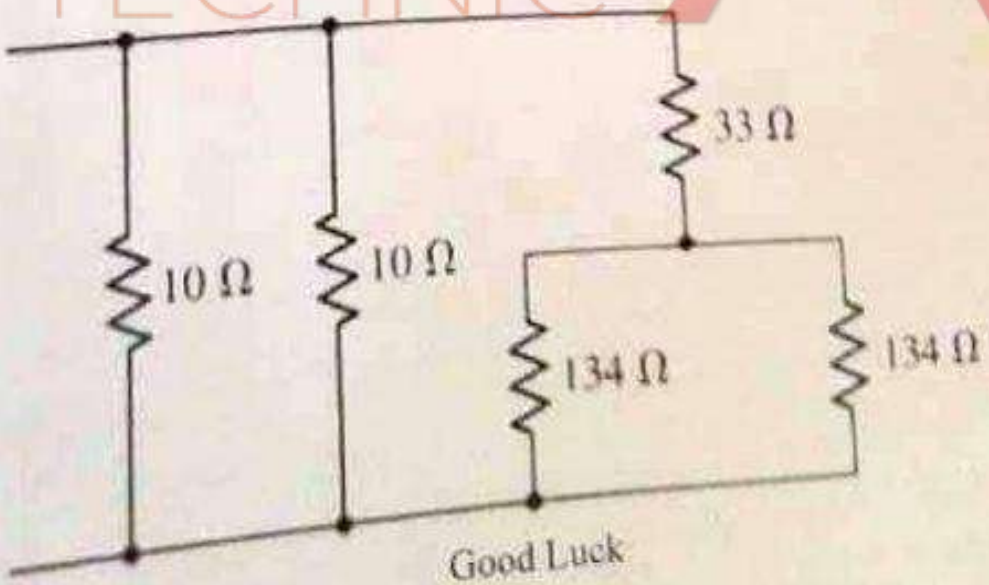
For the single node pair circuit shown in figure determine:

1- The voltage V.
2- The current through 10 Ω resistor
3- The power supplied by 10A current source



Problem 3.[4 points]

Find the equivalent resistance for the network shown in figure



Good Luck

Spark POLYTECHNIC

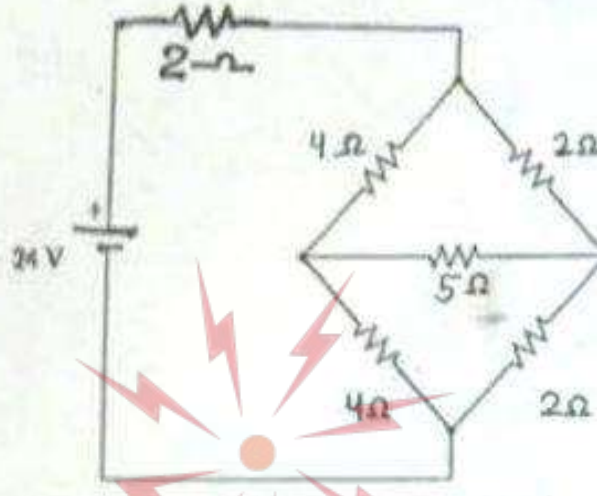
Electric Circuits – *final* Exam – May 20th , 2015

Q1 : For the given circuit find:

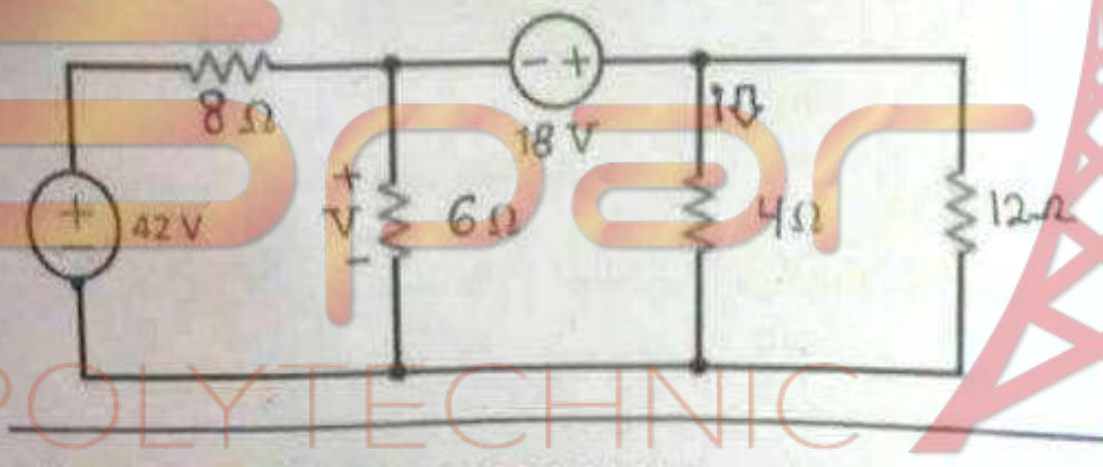
1) The equivalence resistance.

2) The power generated by the source.

3) The voltage across 5Ω resistor.

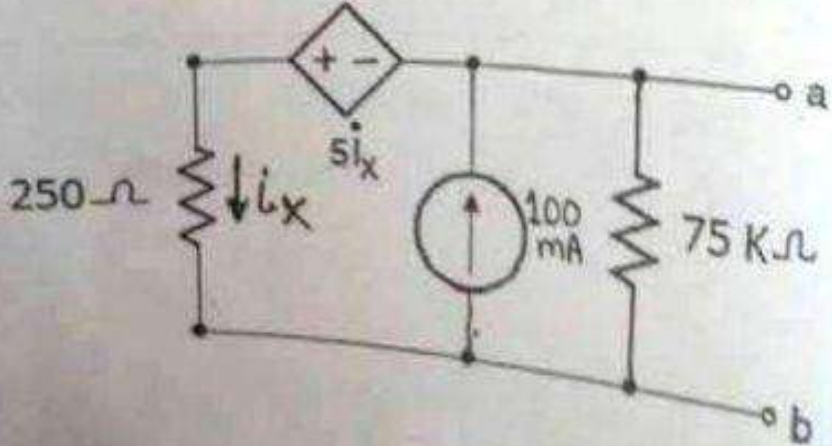


Q2: Using Nodal analysis, find V and i .



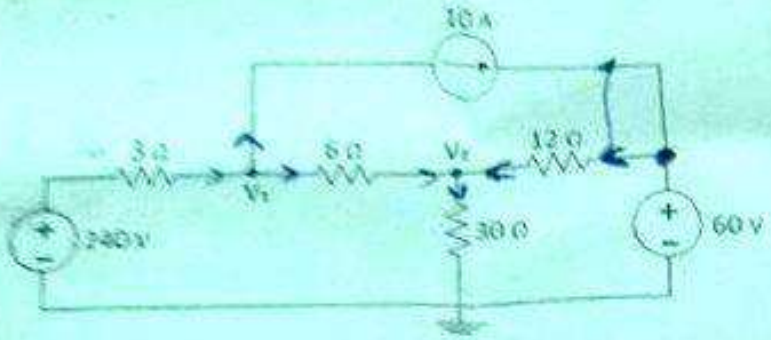
POLYTECHNIC

Q3: Answer the questions next page.



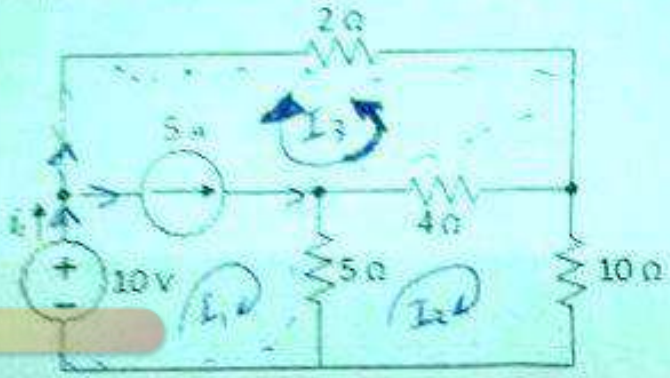
Question One:

Use nodal analysis to determine V_1 and V_2.



Question Two:

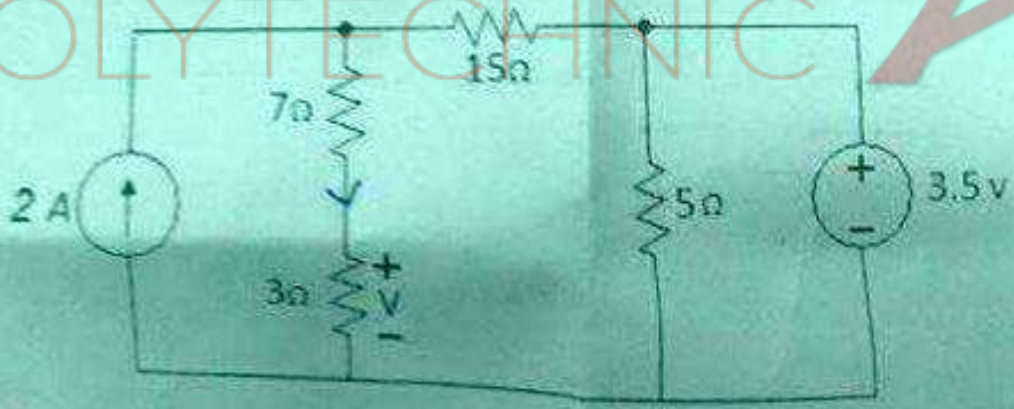
Use mesh analysis to calculate the current of the voltage source i_s.



Question Three:

Use superposition principle to compute the voltage 'V' and the power absorbed by th

5 - Ω resistor.



GOOD LUCK

b) Calculate the power dissipated in the 40 Ω resistor and the voltage labeled Vc for the circuit given in fig 3.b

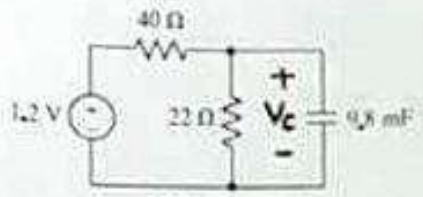
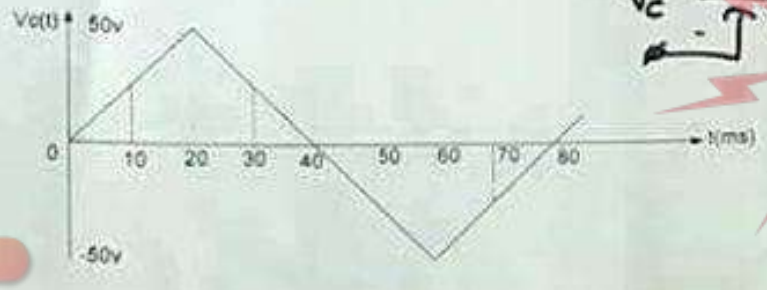


fig 3.b

Q4: The voltage across a 33 mF capacitor is shown graphically in Fig 4 : (8 marks)

a) Assuming the passive sign convention, sketch the resulting current waveform across the capacitor.

b) Find the energy stored in the capacitor at t=20ms.



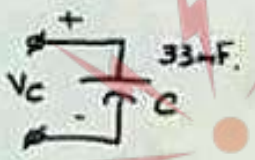


Fig 4

Q5: For the circuit shown in Fig 5 the switch opens at t=0 after having been closed for a long time. Find: (12 marks)

a) $i_1(t)$ for $t >= 0^-$.
b) $i_L(t)$ for $t >= 0^+$.
c) $V_2(t)$ for $t >= 0^+$.

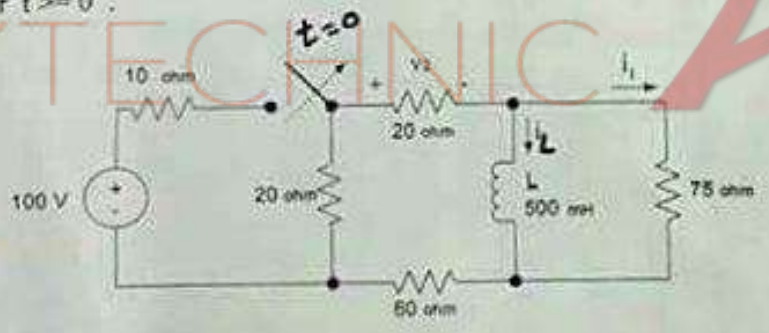


Fig 5

Good Luck